Do Now · retrieval

Lesson 133 · Hashing for Integrity, Passwords and Data Lookup

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

How does hybrid encryption usually protect bulk web traffic?

Q2

What are the outputs of a half adder when A=1 and B=1?

Q3

In Big O notation, what is binary search worst-case complexity?

Q4

When does a linear search stop?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Compute a digest — A cryptographic hash produces a reproducible fingerprint of file content.

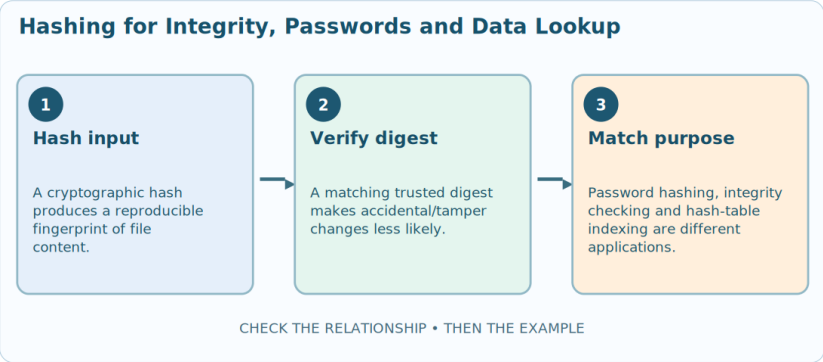
Core idea 2

Compare for integrity — A matching trusted digest makes accidental/tamper changes less likely.

Core idea 3

Choose correct use — Password hashing, integrity checking and hash-table indexing are different applications.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

hash function — Maps input data to a fixed-size or bounded-size result.

digest — Output of a cryptographic hash function.

collision — Two distinct inputs produce the same hash result.

salt — Random value combined with a password before secure password hashing.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A publisher provides a trusted SHA-256 digest of a file. Explain how a pupil can check the download.

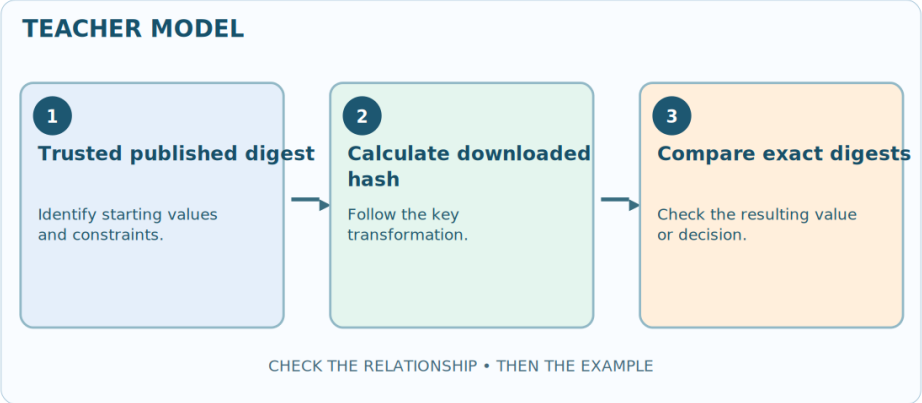
Follow the worked method

1. Trusted published digest — Find the given data and name the required result.

2. Calculate downloaded hash — Explain why this intermediate stage is necessary.

3. Compare exact digests — Calculate SHA-256 digest of received file and compare with the digest from a trusted source. If different, file changed or was corrupted. If equal, this strongly supports matching content, assuming trusted digest and secure algorithm; it does not by itself prove publisher identity.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Choose random salt
Find the given data and name the required result.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State three different uses of hashing with one correct explanation each. [6]

Question 2

Explain what a hash collision is and why it matters. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate the claim 'a hash can decrypt a password and proves a file came from its claimed author'. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A hash output can normally be decrypted to recover the exact original password.

Use key vocabulary from memory

1. Explain hash function in your own words.

2. Explain digest in your own words.

3. Explain collision in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.